4.4 TYPES OF LEVELLING INSTRUMENTS

The following are the different types of levels :

I. DUMPY LEVEL :

A levelling Instrument, also called the level, is used for the determination of levels. The dumpy level is commonly used for levelling is designed by Gravatt. The essential features of the dumpy level are shown in Fig. 4.3. It consists of a telescope which is rigidly fixed to its support. It can neither rotated about its longitudinal axis nor can it be removed from its support. It is very advantages when several observations are to be made with one set-up of the instrument.

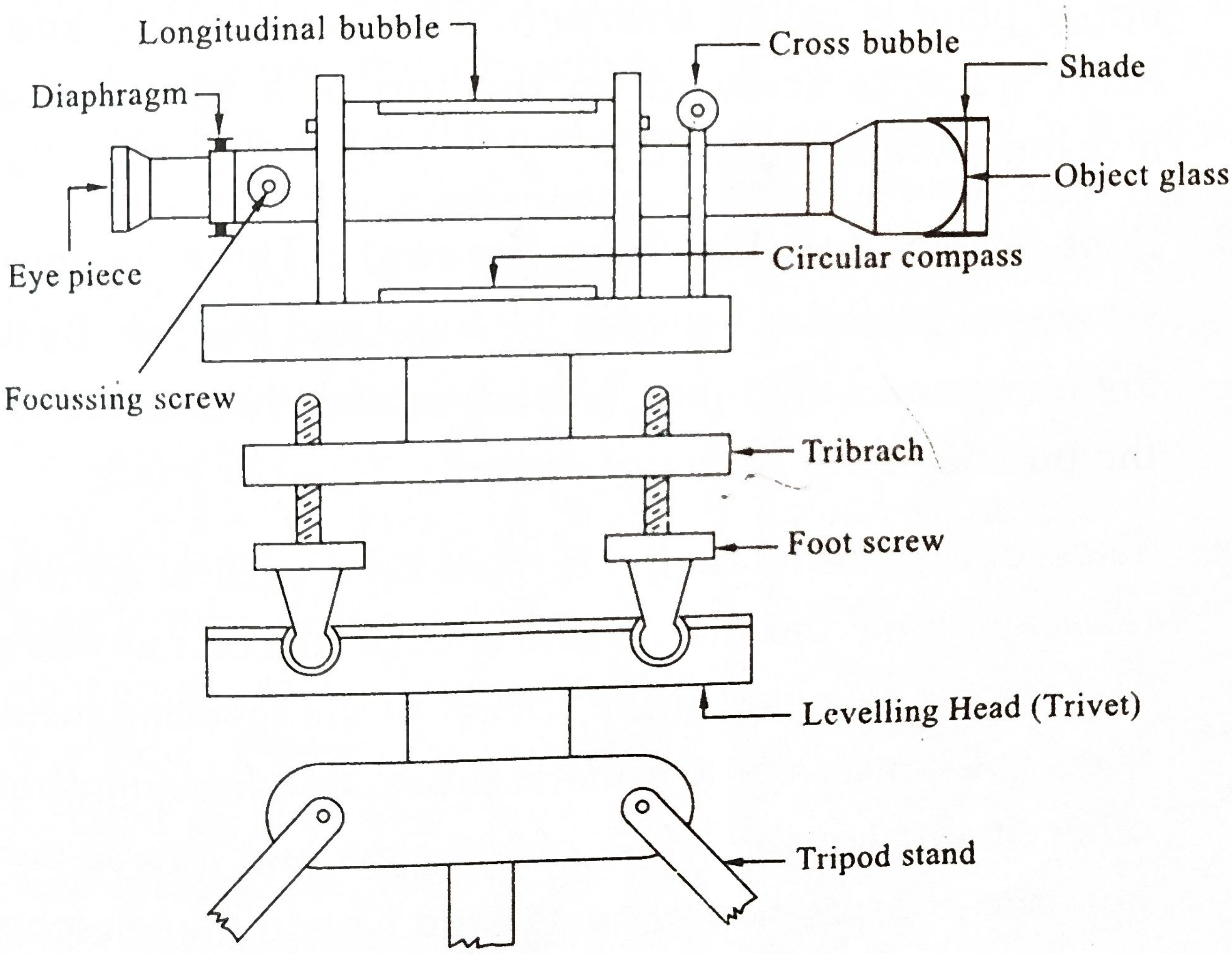


FIG 4.3 :

Description of Dumpy Level :

1. ***Tripod Stand :*** The level is fitted to the tripod stand while taking the levels in the field. It consists of a head and three legs with pointed metal shoes. The tripod head has an external screw to which the lower plate (trivet) of the level is screwed. Tripods are generally made of wood or aluminium. Two types of Tripods are commonly used for levelling. They are :

(a) ***Tripod with Constant Length Legs*** : The fixed-leg tripods provide more rigidity and stability to the levelling instrument, and are, therefore, preferred.

(b) ***Tripod with Extensive Legs*** : This type of tripod provide more flexibility, as the legs can be adjusted to suit any suitable length. And also the extensible leg tripods can be transported easily as they reduce in overall size when folded.

2. **Levelling Head :** The Levelling head consists of two parallel plates held apart by three (or four) levelling screws. The upper plate is called ***tribrach.*** The lower plate, known as trivet stage, is screwed on the top of a tripod when the instrument is to be used.

3. **Foot Screws (or) Levelling Screws)** : Three (or four) foot screws are provided between the trivet and tribrach. By turning the foot screws the tribrach can be raised or lowered to bring the bubble to the centre of its run.

4. **Telescope** : The telescope is fixed on a vertical spindle. The telescope tube and the vertical spindle are cast as one piece. The spindle revolves in the socket of the levelling head. The telescope consists of two metal tubes, one moving within the other. It also consists of an object glass and an eye-piece on opposite ends. A diaphragm is fixed with in the telescope just infront of the eye piece. The diaphragm carries cross-hairs. The telescope is focussed by means of a focussing screw.

   Depending upon the arrangement provided for bringing the image of the object formed by the objective exactly in the plane of cross-hairs, there are two types of telescopes.

   (a) External-focussing telescope.

   (b) Internal-focussing telescope.

(a) ***External-Focussing Telescope*** : In the external focussing telescope, the objective and the eye-piece are mounted on two different tubes so that one can slide inside the other.

The rack and pinion arrangement is provided in the telescope for easy sliding of one tube over the other. The focussing screw is provided on the outer tube. By turning the focussing screw, the objective is moved in or out relative to the diaphragm. Focussing is achieved when a clear image of the object is formed in the plane of the cross-hairs.

(b) ***Internal-Focussing Telescope*** : In internal focussing telescope, the objective and eye-piece are in fixed position. An additional double-concave lens is fitted with rack and pinion arrangement between the diaphragm and the objective. This lens moves to-and-fro when the focussing screw is turned and the image of the object formed by the objective is brought in the plane of cross-hairs of the diaphragm.

5. **Bubble Tubes** : Two bubble tubes, one called the longitudinal bubble tube and other the cross-bubble tube, are placed at right angles to each other on the top of the telescope or on its one side. These tubes contain spirit bubble. The bubble is brought to the centre with the help of the foot screws. An inclined mirror is provided over longitudinal level tube to enable the observer to view the bubble from the eye-piece end of the telescope without moving round the telescope.

6. **Diaphragm** : The surveying telescopes are designed in such a manner, that the image of the object is always formed in a fixed plane, when the instrument has been focussed. The fixed plane at which the image is formed is always made to lie in the plane of the cross-hairs in a ring called the ***diaphragm***.

The diaphragm may be two types. In the first type a brass ring fitted inside the telescope, just infront of the eyepiece, and it carries cross-hairs made of spider webs or platinum wires or silk threads stretched across the ring. The second type of diaphragm consists of a thin glass disc mounted in a circular metal ring and the cross-hairs are marked by fine scratch marks made on the glass disc.

The diaphragm ring is held in position in the telescope by four fine capstan-headed screws arranged at right angles. The ring can be given a small amount of movement by turning capstan screws. Thus it can be adjusted slightly horizontally or vertically, and with the movement of ring the cross-hairs can be adjusted, when required.

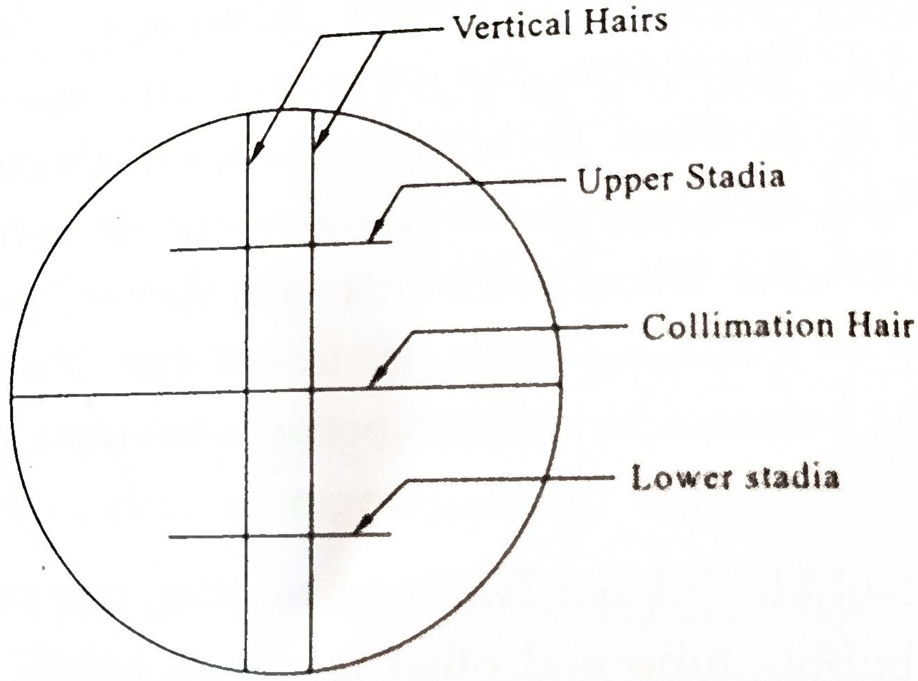


FIG 4.4 :

The Fig. 4.4 shows the general arrangement of cross hairs on a diaphragm. The middle horizontal hair representing the line of collimation. The two vertical hairs are meant for maintaining the verticality of the staff.

It also consists of a two additional horizontal hairs, one above and the other below the central horizontal hairs. These hairs are called stadia hairs, and are used in stadia tacheometry for computing the horizontal and vertical distances from the instrument, to that of the staff.

Compass : A compass is provided below the telescope used for taking the magnetic bearing of a line when required. The compass is graduated in such a way that a ***'pointer'***, which is fixed to the body of compass, indicates a reading of 0° when the telescope is directed along the north line.

In some compasses, the pointer shows a reading of few degrees when the telescope is directed towards the North. This readings should be taken as the initial reading. The bearing is obtained by deducting the initial reading from the final reading of the compass.

II. TILTING LEVEL :

The tilting level is commonly used for precise and quick levelling. It is similar to the dumpy level in many respects, but the telescope of the tilting level is not rigidly fixed to the vertical spindle (Fig. 4.5). The telescope can be tilted on a pivot about horizontal axis in the vertical plane upwards or downwards through a small angle by means of a tilting screw. This tilting screw acts against a spring.

A pond level is fixed to the upper plate (tribrach) of the levelling head for approximate levelling by foot screws. Exact levelling is done using the tilting screw. The tilting screw is turned up or down till the bubble of the level tube attached to the telescope is at the center of the run. At this stage, the line of sight is horizontal. Before taking each staff reading, the bubble is brought to the center of run. The tilting screw is usually graduated, So that gradient lines can also be set out.

Tilting level is more-accurate than dumpy level. It has a shorter telescope and lighter in weight. The tilting arrangement saves time required for temporary adjustment.

In some tilting levels, the conventional three foot screws arrangement is replaced by a ball and socket joint. This enables the level to be quickly levelled, approximately. Exact levelling is done by tilting screw.

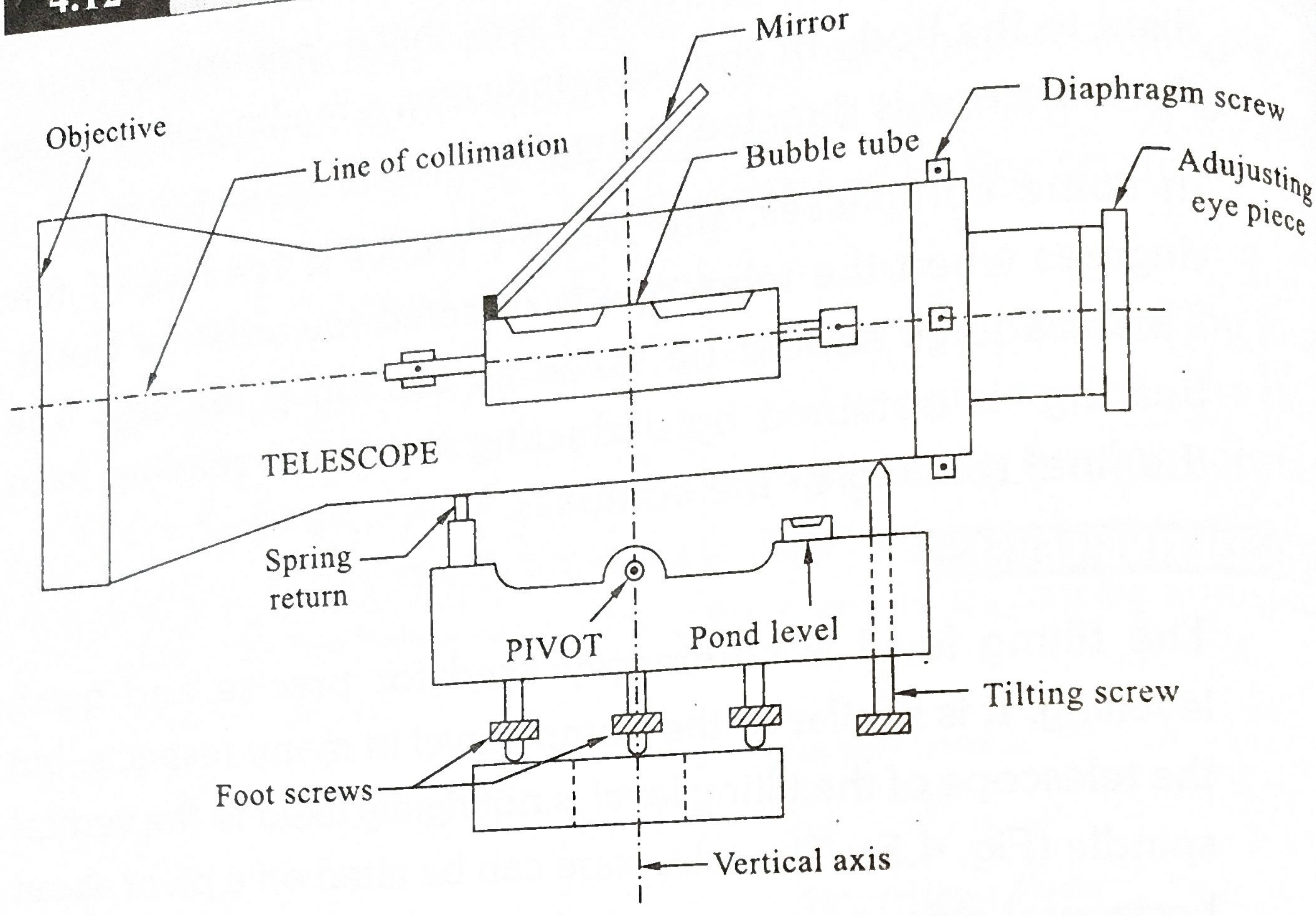
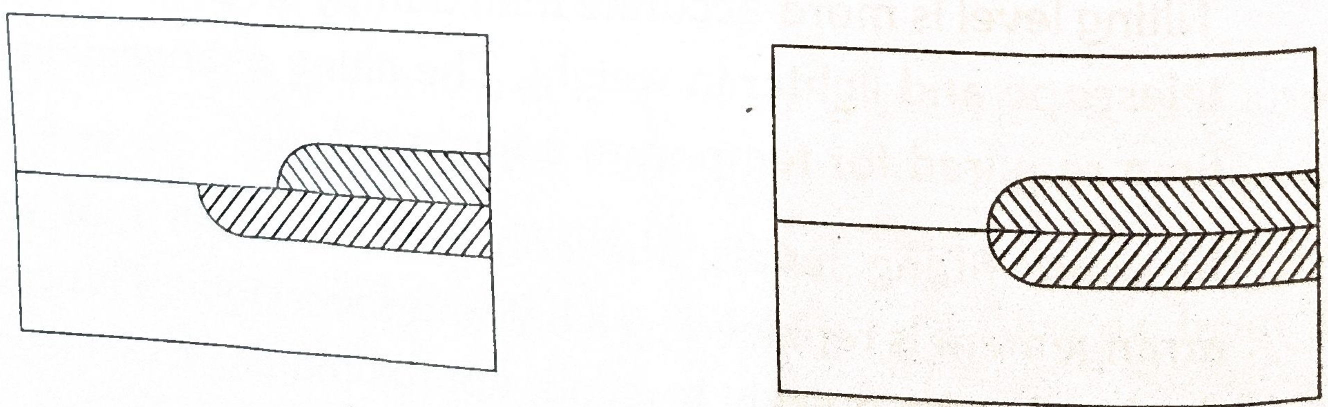


FIG 4.5 : Tilting Level

In some tilting levels, a horizontal graduated circle is provided to measure the horizontal angles. There is a clamping screw and a slow motion tangent screw for clamping the telescope in a particular direction.

In modern tilting levels, the bubble tube is enclosed in a metal cover. An optical arrangement provided at the side of the eye piece enables the two of the bubble to be viewed side by side as shown in Fig. 4.6 (a). The bubble is central when the two end appear to agree as in Fig. 4.6 (b).



(a) (b)

FIG 4.6 :

III. WYE LEVEL :

A wye level consists of a telescope held in two vertical wye supports. The wye supports are two-curved clips which may be raised to enable the telescope to be rotated about its longitudinal axis. It may be removed and turned end to end. (Fig. 4.7).

When the clips are tightened, the telescope is held in supports rigidly. A level tube is attached to the stage carrying the wyes. The levelling head is similar to that of a dumpy level. In some levels, a clamp and fine motion tangent screw, are provided to control the movement in the horizontal plane.

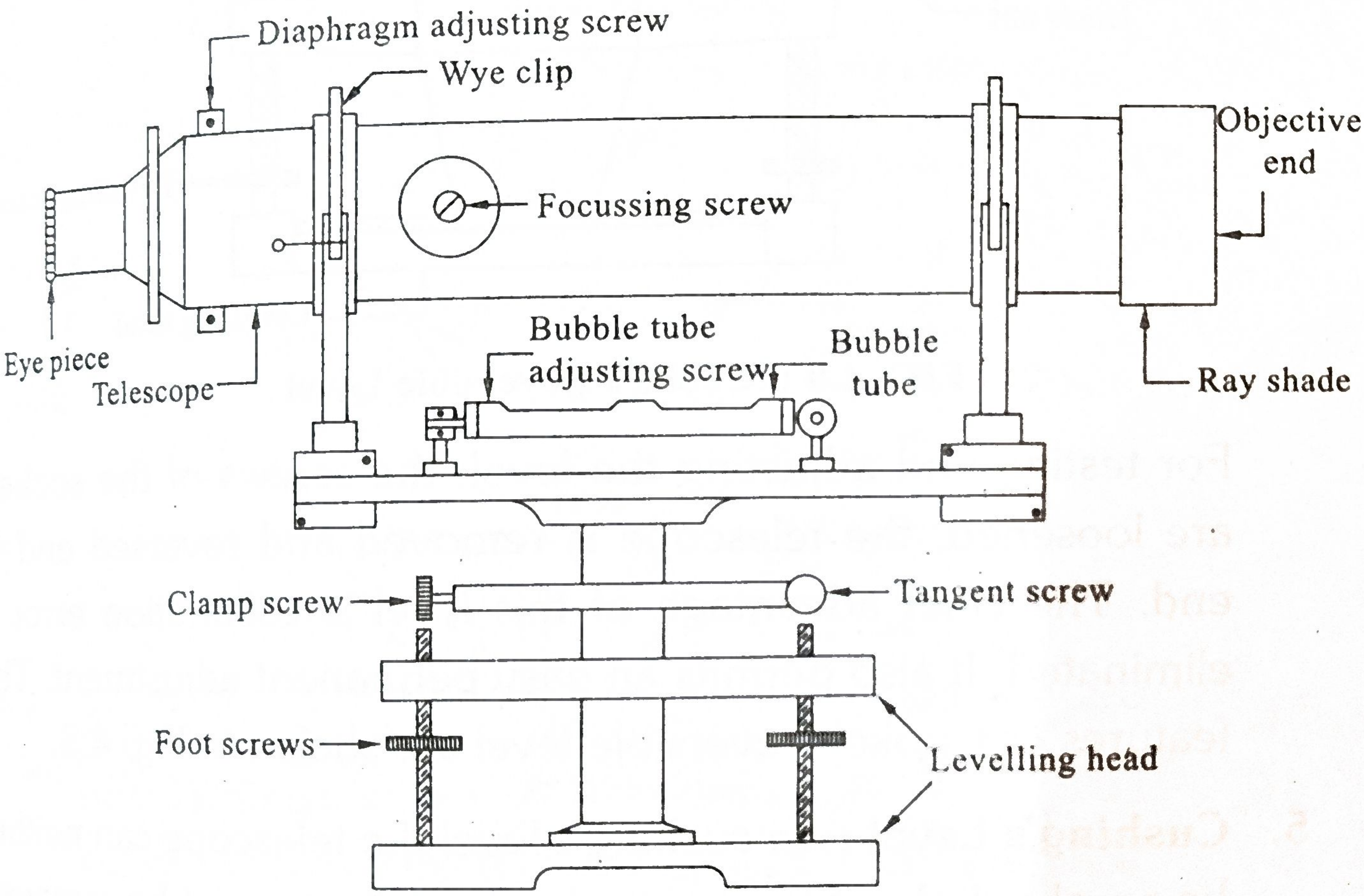


FIG 4.7 : A Wye Level

Wye-level has an advantage over dumpy level is that its adjustments can be tested rapidly. The disadvantage is that it carries many loose and open parts, which are liable to frictional wear.

4. **Cooke's Reversible Level :** In this level, the features of both dumpy and wye levels are combined, so that the advantages of each could be utilized. The telescope is supported on two rigid sockets. The telescope is introduced from either end and then fixed in position by means of screws. The sockets are

rigidly connected to the spindle through a stage. Once the telescope is introduced into sockets and screws tightened, it acts as a dumpy level.

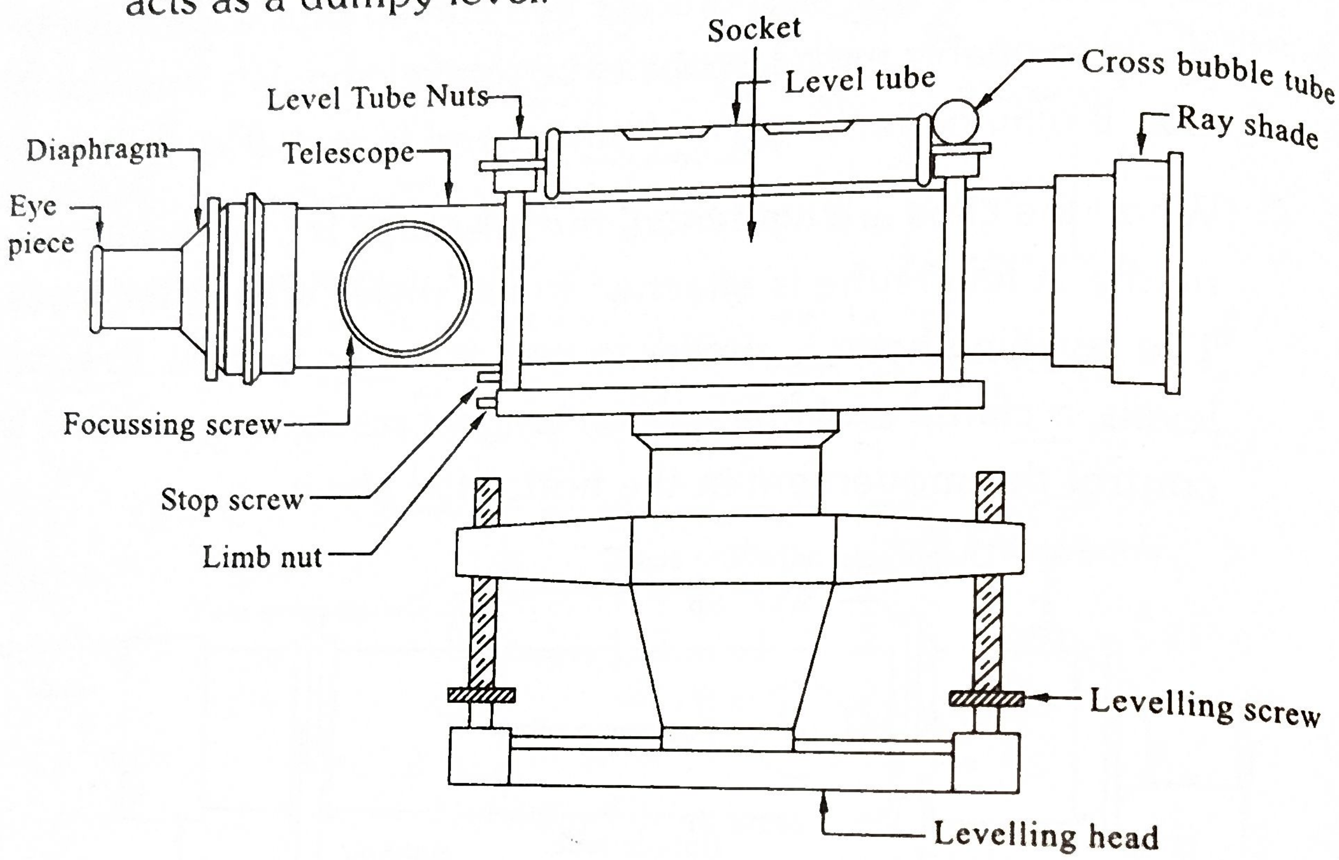
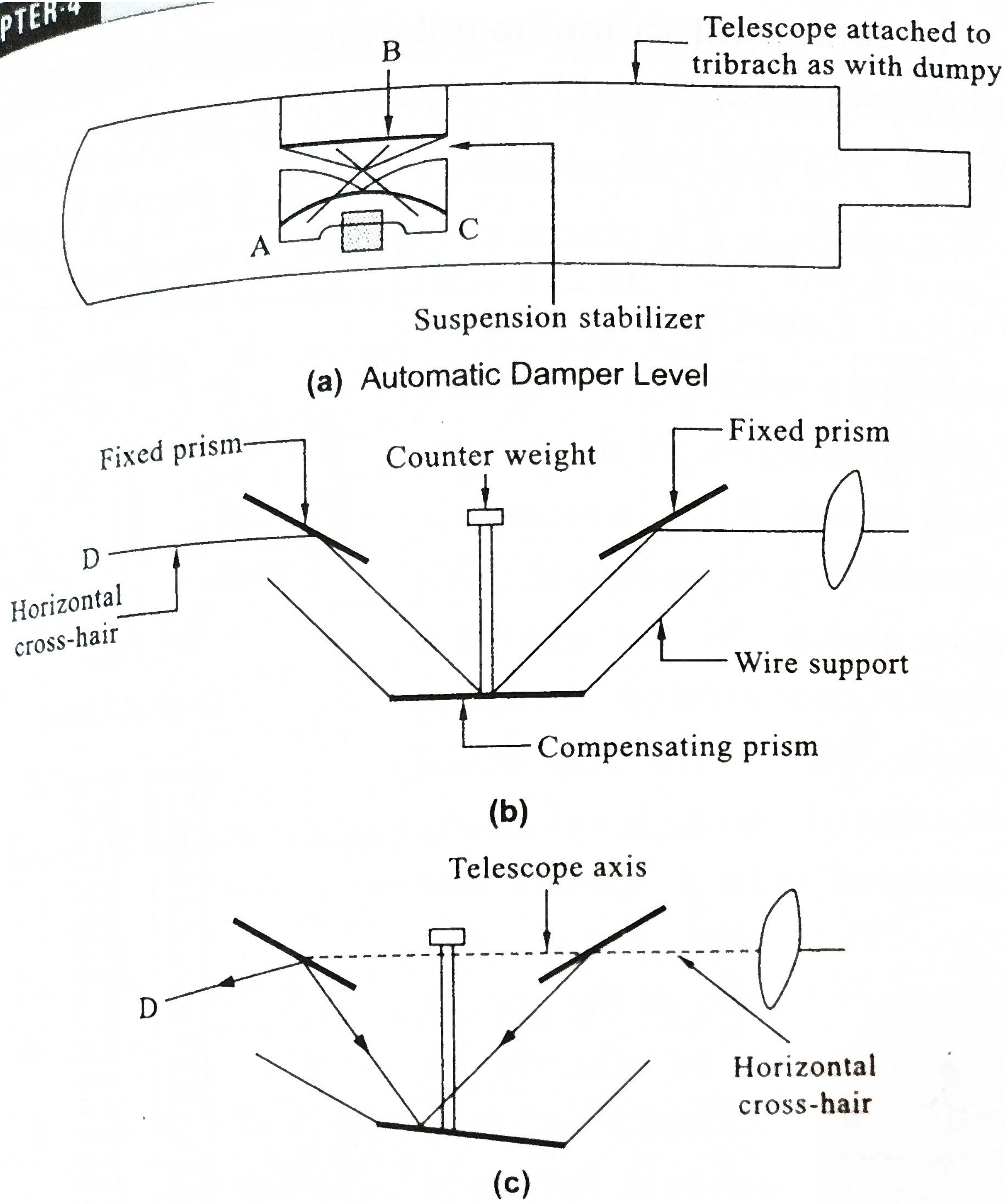


FIG 4.8 : Cooke's Reversible Level

For testing and adjusting the level, the screws of the sockets are loosened, the telescope is removed and reversed end to end. The chief advantage of this level is collimation error is eliminated. It also permits an easy permanent adjustment. The features of Cooke's reversible level are shown in Fig. 4.8.

5. **Cushing's Level :** In cushing's level the telescope can neither be revolved about its longitudinal axis nor can it be removed from its socket. However, the eye-piece and object glass are removable and can be interchanged from end of the telescope to the other end..

6. **Automatic (or) Adjusting Level :** It is similar to the dumpy level with its telescope fixed to the tribrach. For approximate levelling a circular spirit bubble is attached to the side of the telescope. For accurate levelling a stabilizer (Fig. 4.9 [*a*]) is fitted inside the telescope, which automatically levels the instrument. The automatic levelling utilizes the action of gravity in its operation.



(a) Automatic Damper Level

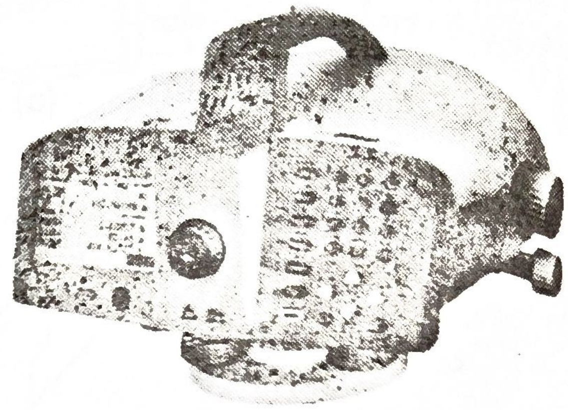
(b)

(c)

FIG 4.9 : Automatic Level

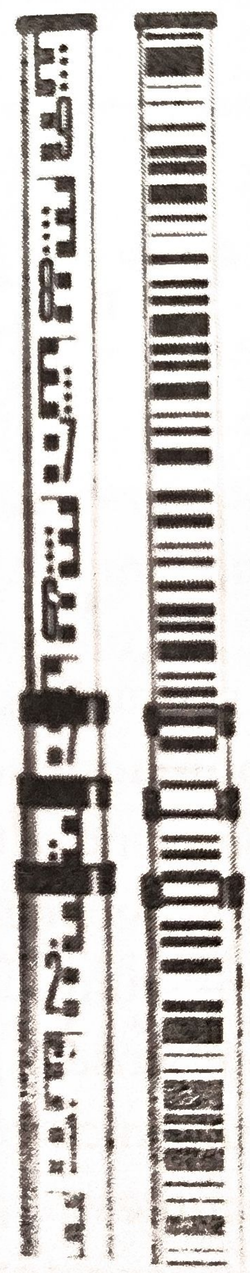
A prismatic device called compensator is suspended on fine nonmagnetic wires. A compensator is an optical system consisting of two fixed prisms (Fig. 4.9 (b)) placed in the optical path between the eyepiece and the objective. An inverted pendulum supported by four non-magnetic wires operates the compensating prism that keeps the image of the sighted point at the intersection of the cross-hairs at *D*. When the instrument becomes approximately level, the action of gravity on the compensator causes the optical system to swing into the position that provides a horizontal line of sight. The pendulum moves until its centre of gravity is over the intersection of the lines of the wire supports. This moves the compensating prism so that the horizontal line of sight is brought to the horizontal cross-hairs at *D*. (Fig. 4.9 (c)).

It is much simpler to use as it gives an erect image. It is very rapid in operation. There is no chance of errors due to bubble setting. The main disadvantage is that it cannot be used in strong winds and at places suspectible to vibrations.



Digital Level

Digital Levelling : Digital levelling is a relatively new system of determining height differences using near fully automatic instruments and methods. This accomplished by the use of a pattern recognition imaging system built into the level instrument and level rods graduated with a special bar code. The user points the instrument at one of the special rods, focuses as clearly as possible, and presses a button to take the measurement. An image of the barcode rod is received at the instrument and is correlated to an internal digital image of the rod. This allows the internal software to determine where the level line of sight is intercepting the rod. The height above the footplate or zero point is then computed along with the distance to the rod. These measurements are displayed and then used to compute the difference of elevation between the backsight and foresight rods. Apart from this automation, leveling with a digital level is much the same as with an optical level. In fact most, digital levels have cross-hairs and can be used optically.



Bar Coded Staff